

A $2p - 3$ Upper Bound for a Bounded-Exponent Refinement of Erdős Problem 246

Zhao Song*

Song Yue

September 3, 2026

Abstract

Let $p, q > 1$ be coprime integers, and let $K(p, q)$ be the least nonnegative integer K such that every sufficiently large positive integer is a sum of distinct terms from $Y_K(p, q) := \{p^a q^b : a \geq 0, 0 \leq b \leq K\}$. Erdős Problem 246 concerns completeness when both exponents are unrestricted [Erd61]; Davenport's bounded-exponent refinement asserts that $K(p, q)$ is finite [Bir59]. We prove $K(2, q) = 0$ and $K(p, q) \leq 2p - 3$ for $p \geq 3$. This improves the bound $K(p, q) \leq 4p - 5$ of Bergelson and Simmons [BS17, Corollary 1.12].

1 Introduction

Let $p, q > 1$ be coprime integers. A set of positive integers is *complete* if every sufficiently large positive integer is a sum of distinct elements of the set. Erdős [Erd61] asked whether $\{p^a q^b : a, b \geq 0\}$ is complete; this is Erdős Problem 246. Birch [Bir59] proved completeness. Davenport observed, as recorded by Birch [Bir59], that only finitely many q -levels are needed. For a nonnegative integer K , define $Y_K(p, q) := \{p^a q^b : a \geq 0, 0 \leq b \leq K\}$. Set $K(p, q) := \min\{K \in \mathbb{Z}_{\geq 0} : Y_K(p, q) \text{ is complete}\}$. We call b the q -level of a term $p^a q^b$.

Hegyvári [Heg00] obtained the first effective bound for $K(p, q)$. Fang [Fan11] and Chen and Fang [CF12] successively reduced its exponential height. Fang and Chen [FC17] later proved the triply exponential estimate $K(p, q) < 2^{2^{q^{2p}}}$.

Bergelson and Simmons [BS17, Corollary 1.12] introduced a different completeness criterion and proved $K(p, q) \leq 4p - 5$ for all coprime $p, q > 1$, replacing the earlier exponential estimates by a bound linear in p .

The known lower bounds are logarithmic in general and linear on a congruence family. Chen, Fang, and Hegyvári [CFH16, Theorem 1.1(ii)] proved $K(p, q) \geq \lceil \log_2 p \rceil - 1$ for every q coprime to p . Xue, Fang, and Ma [XFM23] obtained the stronger estimate $K(p, q) \geq p - 1$ when $p \geq 3$ and $q \equiv 1 \pmod{p}$.

Mixed-power representations also arise in computer arithmetic. The double-base number system represents integers as sums of mixed powers $2^a 3^b$. Dimitrov, Jullien, and Miller [DJM98] use such representations for modular exponentiation. Dimitrov, Imbert, and Mishra [DIM05] use signed double-base chains for elliptic-curve scalar multiplication. These applications motivate the study of integer representations with restricted summands. In Davenport's refinement, grouping summands with the same p -exponent gives a base- p expansion whose digits are subset sums of $1, q, \dots, q^K$. Bounds for $K(p, q)$ therefore control the number of q -levels needed for these nonnegative expansions of every sufficiently large integer.

*magic.linuxkde@gmail.com.

1.1 Our result

Table 1: Evolution of bounds for Davenport’s bounded-exponent refinement of Erdős Problem 246. Panel (a) records upper bounds, and panel (b) records lower bounds. Here $d := 1152(\log_2 p)(\log_2 q)$ and $D := 2^{q^{2p+3}}$.

(a) Upper bounds			
Year	Authors	Reference	Bound for $K(p, q)$
1959	Birch; Davenport	[Bir59]	finite, non-effective
2000	Hegvári	[Heg00]	effective, fourfold exponential
2011	Fang	[Fan11]	p^{d^D}
2012	Chen–Fang	[CF12]	d^D
2017	Fang–Chen	[FC17]	$< 2^{2^{q^{2p}}}$
2017	Bergelson–Simmons	[BS17]	$4p - 5$
2026	This paper	Theorem 4.4	$2p - 3$

(b) Lower bounds			
Year	Authors	Reference	Bound for $K(p, q)$
2016	Chen–Fang–Hegvári	[CFH16]	$\lceil \log_2 p \rceil - 1$ for every q coprime to p
2023	Xue–Fang–Ma	[XFM23]	$p - 1$ if $p \geq 3$, $q \equiv 1 \pmod{p}$

We state our result as follows.

Theorem 1.1 (Main upper bound, informal version of Theorem 4.4). *Suppose that $p, q > 1$ are coprime integers. Then*

$$K(p, q) \leq \begin{cases} 0, & p = 2, \\ 2p - 3, & p \geq 3. \end{cases}$$

For $p \geq 3$, the theorem improves the Bergelson–Simmons bound [BS17, Corollary 1.12] from $4p - 5$ to $2p - 3$, saving $2p - 2$ q -levels.

Roadmap. Section 2 introduces the earlier arguments and explains our two-block proof. Section 3 gives the notation and auxiliary results. Section 4 proves the upper bound.

2 Technical overview

Section 2.1 introduces Birch’s subset-sum construction and Davenport’s bounded-exponent observation. Section 2.2 explains how Hegvári made that observation effective, at the cost of a fourfold-exponential bound. Section 2.3 introduces the step that Fang removed from Hegvári’s construction, and Section 2.4 presents the second reduction of Chen and Fang. These refinements substantially lower the exponent towers but retain the same constructive framework. Section 2.5 describes the four-component completeness criterion of Bergelson and Simmons and explains the endpoint $4p - 5$. Section 2.6 explains our two-block argument for the bound $2p - 3$.

2.1 Birch’s qualitative argument

The lattice model. Birch [Bir59] represents the term $p^a q^b$ by the lattice point (a, b) . A finite subset of the first quadrant then represents a sum of distinct mixed powers. A counting argument

produces two disjoint finite lattice sets with the same sum. A second observation places a mixed power in every sufficiently large multiplicative interval. Translating a hypothetical late gap by such a mixed power moves it to an earlier gap, so the gaps between the ordered finite subset sums are uniformly bounded.

Birch uses the equal-sum relation at successively separated lattice scales. After translating the two sides and removing an extremal monomial, the relation becomes a signed identity of value one. Disjoint copies of these unit identities fill the bounded gaps and produce an infinite arithmetic progression of finite subset sums. Powers on the coordinate axes then supply the residue corrections between consecutive terms of that progression.

The bounded-exponent observation. Davenport observed that the q -exponents in Birch’s construction can be bounded independently of the integer being represented. Birch [Bir59] states that this follows from the same argument, but omits the quantitative details. The observation proves that $K(p, q)$ is finite without giving an effective strip width.

2.2 Hegyvári’s effective bound

Effective bounded-exponent construction. Hegyvári [Heg00] gave the first effective form of Davenport’s observation. A pigeonhole argument first produces two disjoint representations of the same integer. Removing one summand gives a pair of represented sums whose difference is a mixed power. The construction repeats this procedure at separated exponent scales and rescales the resulting pairs to a common difference. Choosing one member from each pair then produces a long arithmetic progression of subset sums. The disjointness of the representations ensures that every term is used at most once. Each stage also gives an explicit bound on the largest q -exponent required.

Parameter growth. Keeping the representations disjoint requires recursively enlarging the exponent scales. The finite progression must also be long enough to fill the gaps between subset sums of a separate family of mixed powers. Their combination gives an infinite arithmetic progression, after which a further construction covers the residue classes modulo its common difference. The parameters for this last construction depend on the exponent bounds already used to obtain the progression. These successive choices produce a fourfold-exponential upper bound for $K(p, q)$. The resulting cutoff depends only on p and q , so one fixed range of q -exponents represents every sufficiently large integer.

2.3 Fang’s first reduction

Disjoint subset sums and arithmetic progressions. Fang [Fan11] retains Hegyvári’s disjoint-pair construction and its progression-building stage. The finite progression uses even q -exponents, while a separate family with odd q -exponents provides bounded subset-sum gaps. The progression is chosen long enough to fill those gaps, and their combination contains an infinite arithmetic progression. The parity separation keeps the underlying summands disjoint. This still represents only one residue class modulo the common difference, so the final stage must supply corrections for all remaining classes.

Residue completion via Vu’s theorem. Fang [Fan11, Lemmas 2.6 and 2.7] applies Vu’s theorem: for a positive integer m , subset sums of any multiset of m integers coprime to m cover every residue class modulo m . For a progression difference $p^a q^b$ with positive integers a, b , the

powers $q, q^2, \dots, q^{p^a}$ therefore supply every residue modulo p^a . Distinct powers of p divisible by p^a then correct the remaining congruence modulo q^b . This reduces the residue-completion stage of Hegyvári's construction. With $d := 1152(\log_2 p)(\log_2 q)$ and $D := 2^{q^{2p+3}}$, the resulting estimate is $K(p, q) \leq p^{d^D}$. The largest q -exponent needed for residue completion controls this final bound.

2.4 The Chen–Fang refinement

A further reduction in exponential height. Chen and Fang [CF12] reorganize the same effective construction and remove one further step. With the same d and D , they obtain $K(p, q) \leq d^D$. This removes the outer power with base p and saves another complete level of the tower. The residue correction must make the target integer congruent to a term of the infinite progression. The remaining part is then represented by that progression, with the two representations using disjoint summands. Controlling the exponent range in this completion step is essential: the largest range required by any component determines the bound for $K(p, q)$.

Fang and Chen [FC17] later proved the triply exponential estimate $K(p, q) < 2^{2^{q^{2p}}}$. Their argument refines the progression construction and the subsequent residue corrections. It also gives an explicit threshold beyond which every integer is represented, controlling both the permitted exponents and the onset of completeness. These refinements retain an exponential dependence on the parameters. The linear dependence on p comes from the different completeness criterion of Bergelson and Simmons.

2.5 The Bergelson–Simmons linear bound

The four-component criterion. Bergelson and Simmons [BS17, Main Theorem 2.1] separate completeness into four components B_1, B_2, B_3, C . Each B_i satisfies a bounded-deficit condition that makes its finite-sum set syndetic, meaning that it meets every interval of nonnegative integers of some fixed length. The set C supplies an irrational-divergence condition and covers every residue class modulo every positive integer. Their criterion then shows that $B_1 \cup B_2 \cup B_3 \cup C$ is complete.

For the mixed-power set, the four components correspond to four consecutive blocks of $p - 1$ q -levels:

$$\begin{aligned} C_{\text{BS}} &:= \{p^a q^b : a \geq 0, 0 \leq b \leq p - 2\}, \\ B_1 &:= \{p^a q^b : a \geq 0, p - 1 \leq b \leq 2p - 3\}, \\ B_2 &:= \{p^a q^b : a \geq 0, 2p - 2 \leq b \leq 3p - 4\}, \\ B_3 &:= \{p^a q^b : a \geq 0, 3p - 3 \leq b \leq 4p - 5\}. \end{aligned} \tag{1}$$

The first block contains the $p - 1$ low powers needed for the arithmetic conditions. Each of the other blocks contains $p - 1$ geometric rays and supplies one syndetic component. The last range in Eq. (1) ends at $4p - 5$. This gives $K(p, q) \leq 4p - 5$ by Bergelson and Simmons [BS17, Corollary 1.12].

2.6 Our two-block argument

The two-block decomposition. The Bergelson–Simmons criterion uses its first block for the arithmetic conditions and three further blocks for three syndetic components. In our argument, the finite-sum set of the first block is thick, meaning that it contains arbitrarily long intervals of consecutive integers. Thus one syndetic block suffices. This replaces the four-component criterion of Bergelson and Simmons [BS17, Main Theorem 2.1 and Corollary 1.12] by a direct thick–syndetic sum. For $p \geq 3$, the two blocks are

$$C := \{p^a q^b : a \geq 0, 0 \leq b \leq p - 2\}, \quad B := \{p^a q^b : a \geq 0, p - 1 \leq b \leq 2p - 3\}. \tag{2}$$

The C in Eq. (2), is the same as the C_{BS} in the Eq. (1). The B in Eq. (2), is the same as the B_1 in the Eq. (1). Each range contains $p - 1$ values of b , and their union is $Y_{2p-3}(p, q)$. The two blocks therefore contain $2p - 2$ q -levels. Since the levels are indexed from 0, their largest exponent is $(p - 1) + (p - 1) - 1 = 2p - 3$.

Complete residue digits. The low block begins with the $p - 1$ powers $1, q, \dots, q^{p-2}$. Their subset sums cover every residue class modulo p . Suppose that the first j powers represent a set of residues. Allowing the next power q^j replaces that set by its union with the translate by q^j . Coprimality implies that q^j generates the additive group $\mathbb{Z}/p\mathbb{Z}$. A nonempty proper residue set cannot be invariant under this translation, so the union gains at least one residue unless all p residues are already present. Starting with the single residue 0, the $p - 1$ adjunctions therefore give all residue classes. If $q \equiv 1 \pmod{p}$, subset sums of only $t < p - 1$ powers represent at most the $t + 1$ residues $0, 1, \dots, t$. Thus $p - 1$ is also the sharp length for this uniform residue argument. Lemma 4.1 selects from these subset sums a set E containing one representative of each residue class, with $0, 1 \in E$.

Intervals in radix sumsets. For sets $U, V \subseteq \mathbb{R}$ and $c \in \mathbb{R}$, let $U + V := \{u + v : u \in U, v \in V\}$ and $cU := \{cu : u \in U\}$. For $n \geq 1$, let $E_n := E + pE + \dots + p^{n-1}E$. The digit set defines the one-dimensional self-affine set $T := \{\sum_{j \geq 1} e_j p^{-j} : e_j \in E\}$. The matrix $[p]$ is expanding, $x - p$ is irreducible, and E is a complete set of coset representatives. Moreover, $1 \in E$ makes the lattice generated by E equal to $\mathbb{Z}$. Kirat, Lau, and Rao [KLR04, Proposition 3.1] therefore show that the integer translates of T tile $\mathbb{R}$. The set T is compact, and the Baire category theorem gives an open interval $I := (\alpha, \beta) \subseteq T^\circ$, where T° denotes the interior of T . Iterating the self-affine identity $pT = T + E$ gives $p^n T = E_n + T$. Kirat, Lau, and Rao [KLR04, Lemma 3.4] give $(E_n + T)^\circ \cap \mathbb{Z} \subseteq E_n$. Since $p^n I \subseteq (p^n T)^\circ = (E_n + T)^\circ$, the set $p^n I \cap \mathbb{Z}$ lies in E_n . These integer points form a consecutive interval with at least $p^n(\beta - \alpha) - 1$ elements, and their cardinality tends to infinity. Thus E_n contains intervals of unbounded length, as stated in Lemma 3.8.

Embedding the radix intervals in the low block. For a set S of positive integers, let $\text{FS}_0(S) := \{\sum_{s \in F} s : F \subseteq S \text{ is finite}\}$; the empty sum is 0. Every digit $e \in E$ is a subset sum of $1, q, \dots, q^{p-2}$. In the a th radix position, multiplication by p^a turns this representation into a sum of elements $p^a q^b$ from C . Expanding the digits of an element of E_n therefore gives a sum of terms from C . Coprimality and unique factorization make these terms distinct, so $E_n \subseteq \text{FS}_0(C)$. The intervals in E_n therefore show that $\text{FS}_0(C)$ is thick, as stated in Lemma 4.2.

Syndeticity of the upper-block finite-sum set. The second block consists of the $p - 1$ geometric rays with initial coefficients $q^{p-1}, q^p, \dots, q^{2p-3}$ and common ratio p . Coprimality makes the map indexing these rays injective. Chen, Fang, and Hegyvári [CFH16, Lemma 2.1 and proof of Theorem 1.1(i)] prove a bounded-gap estimate for t rays with common base m when $t \geq m - 1$. Here $m = p$ and $t = p - 1$, so the block meets this threshold exactly. Their estimate bounds consecutive gaps by the $(p - 1)$ st coefficient, namely q^{2p-3} . Lemma 3.7 gives the general statement, and Lemma 4.3 applies it to show that $\text{FS}_0(B)$ is syndetic.

Sums of thick and syndetic sets. Let $A := \text{FS}_0(C)$ and $S := \text{FS}_0(B)$. The set S meets every interval of some fixed length, while A contains an interval J of that length. For every sufficiently large N , the reflected interval $N - J := \{N - j : j \in J\}$ consists of nonnegative integers and meets S . If $s \in S \cap (N - J)$, then $N - s \in J \subseteq A$. Thus $A + S$ contains every sufficiently large integer.

Finite sums of disjoint blocks. The two blocks in Eq. (2) are disjoint. Coprimality and unique factorization show that an element cannot occur in both blocks, because their q -exponent ranges do not overlap. A finite subset of $C \cup B$ therefore splits into independent finite subsets of C and B , which gives

$$\text{FS}_0(C) + \text{FS}_0(B) = \text{FS}_0(C \cup B) = \text{FS}_0(Y_{2p-3}(p, q)).$$

Hence $\text{FS}_0(Y_{2p-3}(p, q))$ contains every sufficiently large integer. Thus $Y_{2p-3}(p, q)$ is complete and $K(p, q) \leq 2p - 3$. This is the $p \geq 3$ case of Theorem 4.4. When $p = 2$, the set $Y_0(2, q) = \{2^a : a \geq 0\}$ already has every nonnegative integer as a finite subset sum by binary expansion, so $K(2, q) = 0$. The thick low block removes two of the three extra Bergelson–Simmons blocks, lowering the endpoint from $4p - 5$ to $2p - 3$ and saving $2p - 2$ q -levels.

3 Preliminaries

Section 3.1 defines the finite-sum notation and the notions of completeness, thickness, and syndeticity. Section 3.2 introduces the bounded-gap estimate used for the upper block and the radix-interval lemma used for the lower block.

3.1 Basic notation

Let $\mathbb{N} := \{1, 2, 3, \dots\}$ and $\mathbb{Z}_{\geq 0} := \{0, 1, 2, \dots\}$. We use $\mathbb{Z}$ and $\mathbb{R}$ for the integers and real numbers. For a finite set A , the symbol $\#A$ denotes its cardinality, and for $A \subseteq \mathbb{R}$, the symbol A° denotes its interior. For integers u, v and a positive integer M , the symbol $\gcd(u, v)$ denotes their greatest common divisor, and $u \equiv v \pmod{M}$ means that M divides $u - v$.

Minkowski addition and scalar dilation describe the radix operations used below.

Definition 3.1 (Minkowski sums and dilations). For $A, B \subseteq \mathbb{R}$ and $c \in \mathbb{R}$, define

$$A + B := \{a + b : a \in A, b \in B\}, \quad cA := \{ca : a \in A\}.$$

Finite subset sums record representations by distinct elements.

Definition 3.2 (Finite-sum sets). For $S \subseteq \mathbb{N}$, define

$$\text{FS}_0(S) := \left\{ \sum_{s \in F} s : F \subseteq S \text{ is finite} \right\}.$$

The empty choice $F = \emptyset$ gives $0 \in \text{FS}_0(S)$.

The bounded-exponent problem asks when finite subset sums contain every sufficiently large positive integer.

Definition 3.3 (Complete sets). A set $S \subseteq \mathbb{N}$ is *complete* if $\text{FS}_0(S)$ contains every sufficiently large positive integer.

The low block produces intervals of arbitrarily large length.

Definition 3.4 (Thick sets). A set $A \subseteq \mathbb{Z}_{\geq 0}$ is *thick* if, for every integer $L \geq 0$, there is an integer $u \geq 0$ such that

$$\{u, u + 1, \dots, u + L\} \subseteq A.$$

The upper block supplies a uniform bound on gaps.

Definition 3.5 (Syndetic sets). A set $A \subseteq \mathbb{Z}_{\geq 0}$ is *syndetic* if there is an integer $L \geq 0$ such that, for every integer $u \geq 0$,

$$\{u, u+1, \dots, u+L\} \cap A \neq \emptyset.$$

Throughout the paper, $p, q > 1$ are coprime integers.

Davenport's refinement restricts the exponent of q while leaving the exponent of p unrestricted.

Definition 3.6 (Bounded-exponent sets). For $K \in \mathbb{Z}_{\geq 0}$, define

$$Y_K(p, q) := \{p^a q^b : a, b \in \mathbb{Z}_{\geq 0}, b \leq K\}.$$

The bounded-exponent parameter is

$$K(p, q) := \min\{K \in \mathbb{Z}_{\geq 0} : Y_K(p, q) \text{ is complete}\}.$$

Birch's theorem and Davenport's bounded-exponent observation show that the minimum in Definition 3.6 exists [Bir59].

3.2 Two bounded-gap and radix facts

Chen, Fang, and Hegyvári [CFH16, Lemma 2.1 and proof of Theorem 1.1(i)] prove the bounded-gap estimate below for an indexed sequence. The injectivity assumption converts their statement into the set notation of Definition 3.2.

Lemma 3.7 (Chen–Fang–Hegyvári bounded-gap estimate). *Let $m \geq 2$ and $t \geq m-1$ be integers, and let $1 \leq a_1 \leq \dots \leq a_t$ be integers. Suppose that the map*

$$(i, j) \mapsto m^i a_j \quad (i \in \mathbb{Z}_{\geq 0}, 1 \leq j \leq t)$$

is injective, and set

$$S := \{m^i a_j : i \in \mathbb{Z}_{\geq 0}, 1 \leq j \leq t\}.$$

Let $x_0 := 0$, and enumerate the positive elements of $\text{FS}_0(S)$ increasingly as $x_1 < x_2 < \dots$. Then

$$x_{k+1} - x_k \leq a_{m-1} \quad (k \in \mathbb{Z}_{\geq 0}).$$

In particular, $\text{FS}_0(S)$ is syndetic in the sense of Definition 3.5.

Proof. The indexed family $(m^i a_j)_{i \geq 0, 1 \leq j \leq t}$ lists the elements of S without repetition. Its indexed finite-sum set is therefore $\text{FS}_0(S)$. Chen, Fang, and Hegyvári [CFH16, Lemma 2.1 and proof of Theorem 1.1(i)], with their base p replaced by m , show that consecutive elements of $\text{FS}_0(S)$ differ by at most a_{m-1} .

The set $\text{FS}_0(S)$ is unbounded because it contains $m^i a_1$ for every $i \in \mathbb{Z}_{\geq 0}$. Given $u \in \mathbb{Z}_{\geq 0}$, choose k such that $x_k \leq u < x_{k+1}$.

We have

$$x_{k+1} - u \leq x_{k+1} - x_k \leq a_{m-1}.$$

The first step follows from $x_k \leq u$. The second step follows from the gap bound.

Hence $x_{k+1} \in \{u, u+1, \dots, u+a_{m-1}\}$, so this interval meets $\text{FS}_0(S)$. Definition 3.5 gives the final assertion with $L = a_{m-1}$. $\square$

Kirat, Lau, and Rao [KLR04, Proposition 3.1, Lemma 3.4, and proof of Theorem 3.5] obtain long digit strings from a self-affine lattice tiling. The next lemma gives the one-dimensional specialization needed here.

Lemma 3.8 (Radix intervals). *Let $m \geq 2$ be an integer, and let $E \subseteq \mathbb{Z}_{\geq 0}$ be a complete set of residue representatives modulo m with $0, 1 \in E$. For each integer $n \geq 1$, set $E_n := E + mE + \cdots + m^{n-1}E$. There is an integer $n_0 \geq 1$ such that, for every integer $n \geq n_0$, the set E_n contains an interval J_n of consecutive integers with $\#J_n \rightarrow \infty$ as $n \rightarrow \infty$.*

Proof. Define

$$T := \left\{ \sum_{j=1}^{\infty} e_j m^{-j} : e_j \in E \right\}.$$

This is the self-affine set $T([m], E)$ in the notation of Kirat, Lau, and Rao. Separating the first digit from the remaining series gives

$$mT = T + E. \quad (3)$$

We next verify the lattice-tiling hypotheses. The matrix $[m]$ is expanding, and its characteristic polynomial $x - m$ is irreducible. Since $1 \in E$, the lattice generated by E is $\mathbb{Z}$, so $([m], E)$ is primitive in the terminology of Kirat, Lau, and Rao. The set E is also a complete set of coset representatives for $\mathbb{Z}/m\mathbb{Z}$. Kirat, Lau, and Rao [KLR04, Proposition 3.1] therefore show that the translates $T + z$, $z \in \mathbb{Z}$, tile $\mathbb{R}$.

Let $M := \max E$. The digit space $E^{\mathbb{N}}$ is compact, and the tail of the defining series after N digits is at most $Mm^{-N}/(m-1)$. The sum map is therefore continuous with image T , so T is compact.

If T° were empty, then every translate $T + z$ would be nowhere dense. The tiling would express $\mathbb{R}$ as a countable union of nowhere dense sets, contrary to the Baire category theorem. Hence $T^\circ \neq \emptyset$. Choose real numbers $\alpha < \beta$ such that $I := (\alpha, \beta) \subseteq T^\circ$.

For each integer $n \geq 1$, iterating Eq. (3) separates the first n digits.

We have

$$m^n T = T + (E + mE + \cdots + m^{n-1}E) = T + E_n = E_n + T. \quad (4)$$

The first step follows by iterating Eq. (3). The second step follows from the definition of E_n . The last step follows from the commutativity of Minkowski addition.

Set $J_n := m^n I \cap \mathbb{Z}$. This set consists of consecutive integers whenever it is nonempty. We compare it with E_n .

We have

$$J_n \subseteq (m^n T)^\circ \cap \mathbb{Z} = (E_n + T)^\circ \cap \mathbb{Z} \subseteq E_n.$$

The first step follows from $I \subseteq T^\circ$ because dilation by m^n preserves interiors. The second step follows from Eq. (4). The last step follows from Kirat, Lau, and Rao [KLR04, Lemma 3.4], applied with $A = [m]$, $D = E$, and $D_n = E_n$.

The interval $m^n I = (m^n \alpha, m^n \beta)$ has length $m^n(\beta - \alpha)$. Every open interval of length L contains at least $L - 1$ integers, so $\#J_n \geq m^n(\beta - \alpha) - 1$. Choose n_0 so that $m^{n_0}(\beta - \alpha) > 1$. Then J_n is nonempty for every $n \geq n_0$, and its cardinality tends to infinity with n . Thus the intervals $J_n \subseteq E_n$ have cardinalities tending to infinity, which proves the lemma. $\square$

4 Proof of the upper bound

Section 4.1 proves that the low-block finite-sum set is thick, and Section 4.2 proves that the upper-block finite-sum set is syndetic. Section 4.3 combines these results to prove the main upper bound.

4.1 A thick low block

The low-block argument requires a digit set for radix p . We first obtain it from subset sums of the $p-1$ powers $1, q, \dots, q^{p-2}$.

Lemma 4.1 (Complete residue digits). *Let $p \geq 3$ and $q > 1$ be coprime integers. Set*

$$D := \left\{ \sum_{b=0}^{p-2} \varepsilon_b q^b : \varepsilon_b \in \{0, 1\} \right\}.$$

Then D contains a complete set E of residue representatives modulo p with $0, 1 \in E$.

Proof. Let $G := \mathbb{Z}/p\mathbb{Z}$, and for $x \in \mathbb{Z}$ define $\bar{x} := x + p\mathbb{Z} \in G$. For $0 \leq j \leq p-1$, set

$$A_j := \left\{ \overline{\sum_{b=0}^{j-1} \varepsilon_b q^b} : \varepsilon_b \in \{0, 1\} \right\} \subseteq G.$$

Then $A_0 = \{\bar{0}\}$. For $0 \leq j \leq p-2$, the choices $\varepsilon_j = 0$ and $\varepsilon_j = 1$ give $A_{j+1} = A_j \cup (A_j + \bar{q}^j)$. In particular, $A_j \subseteq A_{j+1}$.

Fix j with $0 \leq j \leq p-2$. Since $\gcd(q, p) = 1$, the element $\bar{q}^j$ generates the additive group G . Indeed, $p\bar{q}^j = \bar{0}$, while $k\bar{q}^j = \bar{0}$ implies $p \mid kq^j$ and hence $p \mid k$.

If $A_j + \bar{q}^j = A_j$, then iteration makes the nonempty set A_j invariant under every translation in G , which forces $A_j = G$. Thus, when $A_j \neq G$, the translate $A_j + \bar{q}^j$ differs from A_j . Translation preserves cardinality, so the union $A_j \cup (A_j + \bar{q}^j)$ contains at least one new element. Therefore $\#A_{j+1} \geq \#A_j + 1$ whenever $A_j \neq G$.

Suppose that $A_{p-1} \neq G$. Since $A_0 \subseteq A_1 \subseteq \dots \subseteq A_{p-1}$, every A_j is proper. Applying $\#A_{j+1} \geq \#A_j + 1$ for $j = 0, 1, \dots, p-2$ gives $\#A_{p-1} \geq \#A_0 + p-1 = p$, which is impossible for a proper subset of the p -element group G . Hence $A_{p-1} = G$.

By the definitions of D and A_{p-1} , reduction modulo p maps D onto G . Choose one element of D above each residue class. Since $0, 1 \in D$, choose 0 and 1 as the representatives of $\bar{0}$ and $\bar{1}$, respectively. The resulting set E is a complete set of residue representatives modulo p and contains 0 and 1. $\square$

We now lift the residue digits through powers of p . The radix-interval lemma then produces arbitrarily long intervals in the low-block finite-sum set.

Lemma 4.2 (Thickness of the low block). *Let $p \geq 3$ and $q > 1$ be coprime integers, and set $C := \{p^a q^b : a \geq 0, 0 \leq b \leq p-2\}$. Then $\text{FS}_0(C)$ is thick.*

Proof. Let E be the set from Lemma 4.1. For each $e \in E$, choose $F_e \subseteq \{0, 1, \dots, p-2\}$ such that $e = \sum_{b \in F_e} q^b$. For $n \geq 1$, set $E_n := E + pE + \dots + p^{n-1}E$. Fix $n \geq 1$ and $N \in E_n$. By the definition of E_n , choose $e_0, e_1, \dots, e_{n-1} \in E$.

We have

$$N = \sum_{a=0}^{n-1} p^a e_a = \sum_{a=0}^{n-1} \sum_{b \in F_{e_a}} p^a q^b.$$

The first step follows from the definition of E_n . The last step follows from $e_a = \sum_{b \in F_{e_a}} q^b$ for every $0 \leq a \leq n-1$.

Every summand belongs to C . The summands are distinct: if $p^a q^b = p^{a'} q^{b'}$, then $\gcd(p, q) = 1$ makes the prime supports of p and q disjoint, so unique factorization gives $a = a'$ and $b = b'$.

Thus N is a sum of distinct elements of C . Definition 3.2 gives $N \in \text{FS}_0(C)$. Since n and N were arbitrary,

$$E_n \subseteq \text{FS}_0(C) \quad (n \geq 1). \quad (5)$$

By Lemma 4.1, the set $E \subseteq \mathbb{Z}_{\geq 0}$ is a complete set of residue representatives modulo p with $0, 1 \in E$. Lemma 3.8 therefore applies with $m = p$ and supplies intervals $J_n \subseteq E_n$ with $\#J_n \rightarrow \infty$. Fix $L \geq 0$, choose n such that $\#J_n \geq L + 1$, and choose an interval of $L + 1$ consecutive integers in J_n . Eq. (5) places this interval in $\text{FS}_0(C)$. Since $L \geq 0$ was arbitrary, $\text{FS}_0(C)$ is thick by Definition 3.4. $\square$

4.2 A syndetic upper block

The upper block consists of $p - 1$ consecutive q -levels. The Chen–Fang–Hegvari estimate controls the gaps between its finite subset sums.

Lemma 4.3 (Syndeticity of the upper block). *Let $p, q > 1$ be coprime integers, and define*

$$B := \{p^a q^b : a \geq 0, p - 1 \leq b \leq 2p - 3\}.$$

Then $\text{FS}_0(B)$ is syndetic.

Proof. For $1 \leq j \leq p - 1$, set $a_j := q^{p-2+j}$. Then $1 \leq a_1 < \dots < a_{p-1}$ and

$$B = \{p^i a_j : i \geq 0, 1 \leq j \leq p - 1\}.$$

The map $(i, j) \mapsto p^i a_j$ is injective. Indeed, if $p^i q^{p-2+j} = p^{i'} q^{p-2+j'}$, then coprimality and unique factorization give $i = i'$ and $j = j'$. Thus all the hypotheses of Lemma 3.7 hold with $m = p$ and $t = p - 1$. Consecutive elements of $\text{FS}_0(B)$ therefore differ by at most $a_{p-1} = q^{2p-3}$. By Definition 3.5, $\text{FS}_0(B)$ is syndetic. $\square$

4.3 Putting it all together

The low-block and upper-block lemmas give the required thick–syndetic decomposition for $p \geq 3$. The remaining case $p = 2$ follows from binary expansion.

Theorem 4.4 (Main upper bound, formal version of Theorem 1.1). *Suppose that $p, q > 1$ are coprime integers. Then*

$$K(p, q) \leq \begin{cases} 0, & p = 2, \\ 2p - 3, & p \geq 3. \end{cases}$$

Proof. If $p = 2$, then $Y_0(2, q) = \{2^a : a \geq 0\}$. Binary expansion gives $\text{FS}_0(Y_0(2, q)) = \mathbb{Z}_{\geq 0}$. Definitions 3.3 and 3.6 therefore give $K(2, q) = 0$.

Assume that $p \geq 3$, and define the two blocks.

$$C := \{p^a q^b : a \geq 0, 0 \leq b \leq p - 2\}, \quad B := \{p^a q^b : a \geq 0, p - 1 \leq b \leq 2p - 3\}. \quad (6)$$

If $x \in C \cap B$, then $x = p^a q^b = p^{a'} q^{b'}$ for exponents with $0 \leq b \leq p - 2$ and $p - 1 \leq b' \leq 2p - 3$. Coprimality and unique factorization give $b = b'$, a contradiction. Thus $C \cap B = \emptyset$.

Define their subset-sum sets.

$$A := \text{FS}_0(C), \quad S := \text{FS}_0(B). \quad (7)$$

Lemma 4.2 shows that A is thick, and Lemma 4.3 shows that S is syndetic. Definition 3.5 supplies an integer $L \geq 0$ such that every interval of $L + 1$ consecutive nonnegative integers meets S . Definition 3.4 then supplies an integer $u \geq 0$ for which the following interval lies in A .

$$J := \{u, u + 1, \dots, u + L\} \subseteq A. \quad (8)$$

Fix $N \geq u + L$. The interval $\{N - u - L, N - u - L + 1, \dots, N - u\}$ consists of $L + 1$ nonnegative integers, so it contains an element $s \in S$. Eq. (8) then gives $N - s \in J \subseteq A$. Thus $N \in A + S$. It follows that $A + S$ contains every integer at least $u + L$. We now identify this sumset.

We have

$$A + S = \text{FS}_0(C) + \text{FS}_0(B) = \text{FS}_0(C \cup B) = \text{FS}_0(Y_{2p-3}(p, q)). \quad (9)$$

The first step follows from Eq. (7). The second step follows from $C \cap B = \emptyset$ and Definition 3.2. The last step follows from Eq. (6) and Definition 3.6.

Eq. (9) shows that $\text{FS}_0(Y_{2p-3}(p, q))$ contains every sufficiently large positive integer. Definitions 3.3 and 3.6 therefore give $K(p, q) \leq 2p - 3$. This proves the $p \geq 3$ case and hence the theorem. $\square$

GenAI Usage Disclosure

Codex were used to polish language, such as grammar and wording.

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